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RZL-Ax Memory on Package (MoP) Equipment List

Intel Corporation
June 2026

Rev 1

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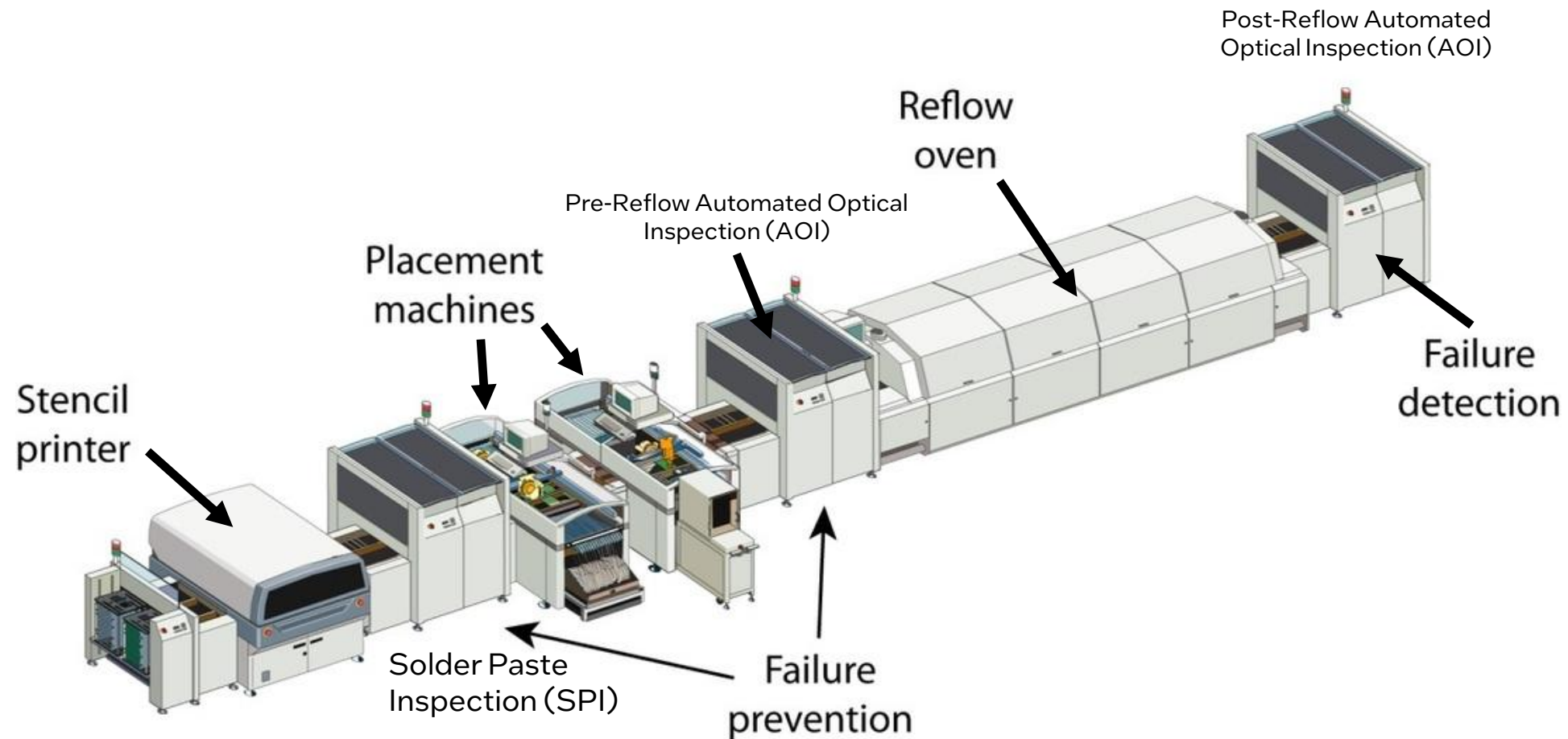
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Revision History

Revision	Changes	Date
Rev 1	Initial Release	June 11, 2026

General SMT Line Layout



To build with MoP (Memory on Package), a general SMT line layout is needed, but with some additional tooling & material additions (see the following pages)

Razor Lake-Ax MoP Equipment List

Placement Machine, with BGA Flux Dip Capability for MoP

Minimum Rqmt's:

- **Pick & Place placement machine:** Accuracy recommendation $\leq 25\mu\text{m}$
- **Fiducials used for memory placement:** Memory can be placed on the SoC either using SoC local or PCB global or fiducials (based on specific equipment capabilities)
- **Memory nozzle size:** Work with your Pick & Place vendor to determine the optimal nozzle diameter to accommodate both memory flux dip & part placement
- **BGA Flux Dip Unit:** In a feeder slot
- **Flux Material:** Example: Shenmao SMBF-08*, a tinted flux (example: black color) may be considered
- **Dip plate cavity coplanarity:** $< 35\mu\text{m}$
- **Wet film gauge:** To check flux height in dip plate cavity

Expected Outcome:

- Align on PoP process settings.
- Successful flux dip process (dip plate, dip dwell, flux height, etc.)

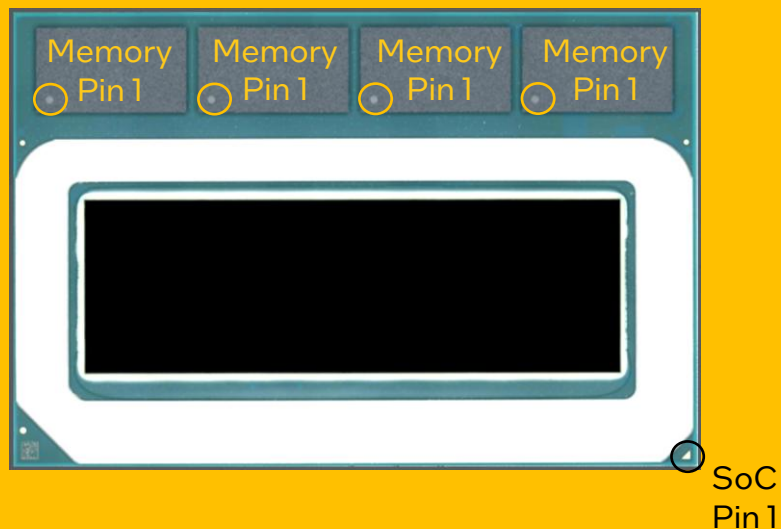
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Razor Lake-Ax MoP Equipment List

Pre-Reflow AOI Inspection for memory

Minimum Rqmt's:

- **Pre-Reflow Automated Optical Inspection (AOI) machine:** To ensure proper memory placement alignment & orientation / polarity
- All 4 RZL-Ax memory devices have the same polarity, as shown below, which the AOI machine should be able to detect.



Expected Outcome:

- Detect if any memory placements are incorrect, either from alignment, orientation or polarity, allowing repairs to be made prior to units entering the reflow oven

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Razor Lake-Ax MoP Equipment List

Reflow Oven, N2
Capable, to solder
MoP on SoC

Minimum Rqmt's:

- **Multi-Zone Reflow oven that is Nitrogen capable:**
Example: O2 $\leq$ 3000 PPM
- **Reflow pallet used:** To prevent motherboard sag or warpage
- **Able to profile with a SAC reflow profile:** As shown below

Attributes	Value
Peak Reflow Temp	235 - 245 °C
Preheat/Rising time (40 - 170 °C)	65 - 90 seconds
Preheat/Rising time (170 - 225 °C)	70 - 100 seconds
Preheat/Rising time (40 - 240 °C)	190 - 240 seconds
Preheat/Rising time (150 - 175 °C)	20 - 90 seconds
Time Above Liquidus (TAL) ($\geq$ 220 °C)	50 - 70 seconds
Max Rising Slope	1 - 3 °C/sec

Expected Outcome:

- Reflow profile successfully developed, meeting key attributes
- O2 levels are met
- Reflow pallet designed to prevent board sag or warpage

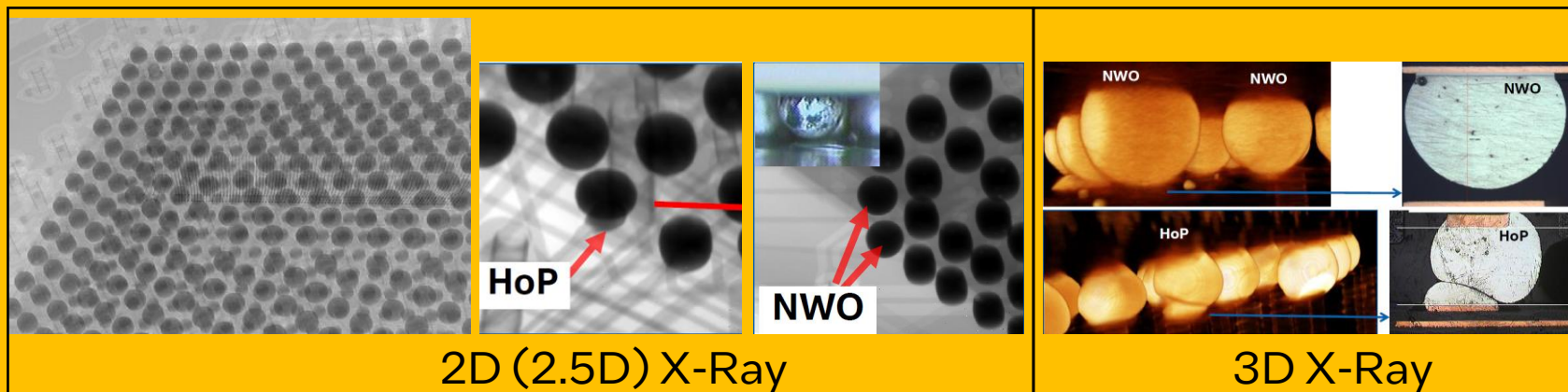
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Razor Lake-Ax MoP Equipment List

X-Ray
Inspection
Tool, to check
MLI solderjoint
quality

Minimum Rqmt's:

- **2D (2.5D) X-Ray inspection tool:** With the ability to tilt or angle the images to inspect the solder joint quality for both the memory (MLI) and SoC (SLI) solderjoints
- **3D X-Ray inspection tool is optional:** But it could be helpful to gain a finer level view of solderjoint quality (example: able to detect SJ open defects)
- **X-Ray Shields and Dosimeters:** To measure & control the amount of radiation (RAD) and dosage exposure into the memories, to protect all memory components from radiation over safe dosage limits
- 2D (2.5D) X-ray should be used 100% during NPI and memory rework activities, with regular sampling during HVM



Note: Generic X-Ray images shown here to demonstrate a few different solder joint view options

Expected Outcome:

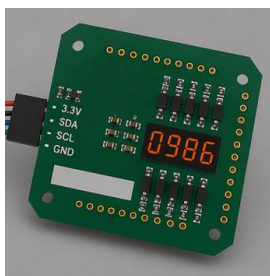
- Able to successfully evaluate Memory SMT yield by X-Ray (X/S and DnP should also be used for NPI builds)

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Razor Lake-Ax MoP Equipment List

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Board Level Testing (Diagnostic)



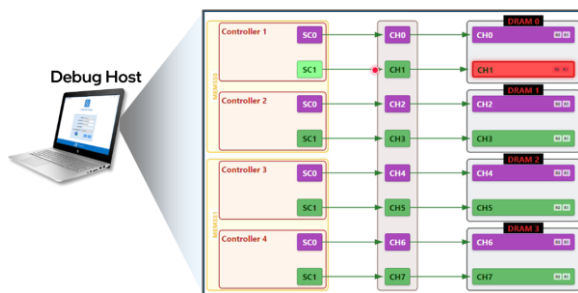
Port 80 Debug Card *1



DCI DBC Cable *1



UART Cable *1



IMDT Software *1

Minimum Rqmt's:

Use case: Manufacturing On-Line Diagnostic or Offline Debug BIOS POST Error Codes (Multi-Phase)

- **BIOS POST Debug Card.**
 - Option 1: Customer legacy debug card (M.2, LPC, PCIE)
 - Requires 4-LED Port 80 Display
 - Option 2: Intel recommended debug card via i2c bus (SDA, SCL)
 - Platform Design Guide for i2c design requirement (*WIP*)
- **Production BIOS to support Post Code Port 80 Multiphase**
 - Firmware codebase will be included as part of the RZL BKC
- **BIOS POST error code decoding table reference**

Use case: Offline Debug

- **Memory Loop Back Display**
 - DDRIO Loopback Firmware Image (Debug BIOS)
 - UART (preferred) or DCI cable- To be enabled in Debug BIOS
 - Debug Host PC with Intel IMDT *2 Software Application installed
- **RMT MRC Serial log and IBECC (In-Band ECC)**
 - Debug BIOS
 - If using DCI: Enable RMT and DCI/DAM, UART disable. Enable IBECC (Error Correction Disabled)
 - If using UART: Enable UART for RMT serial log
 - UART (for RMT MRC serial log) or DCI cable (for RMT or IBECC)
 - Debug Host PC with IMDT Software Application installed

Expected Outcome:

BIOS POST Error Codes Multi-Phase

- Detect memory-related boot failures via Port 80 codes
- Isolate failing memory channel interfaces (e.g., ChA, ChB)
- Display Output: Port 80 LED

Memory Loopback Display

- Enable connectivity test (e.g., Open Solder) for memory DQ/DQS signals
- Display Output: IMDT App GUI

RMT MRC Serial Log

- Check memory margin during initial boot with simplified MRC error log message & force flow test.
- Display Output: IMDT App GUI

IBECC

- Identify functional type failure (i.e., BSOD, System Hang) related to memory corruption.
- Display Output: IMDT App GUI

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*2 IMDT: Intel Memory Diagnostic Tool.

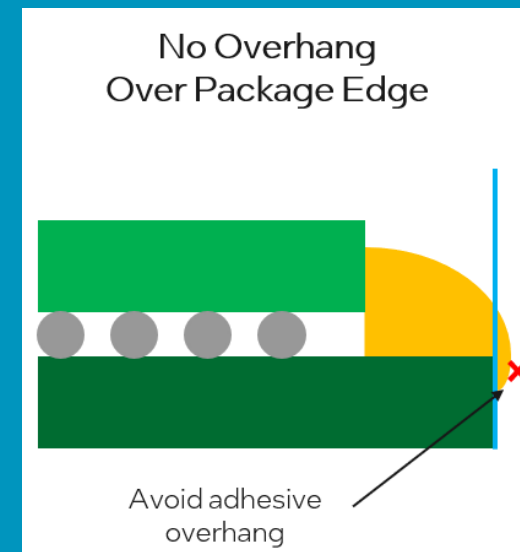
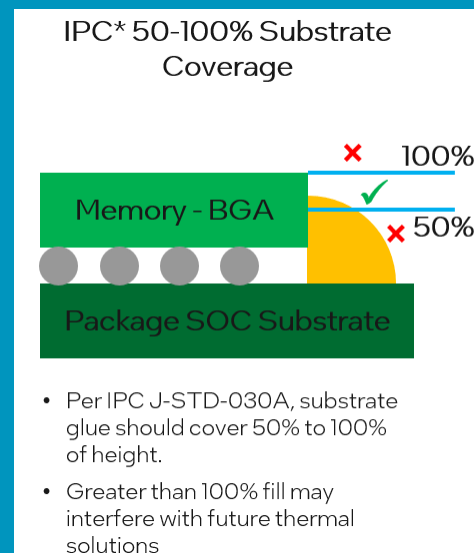
Razor Lake-Ax MoP Equipment List

Edge Bond Adhesive Jet Dispense Equipment

Minimum Rqmt's:

- Recommend glue is dispensed only after the motherboard has passed functional testing (to avoid unnecessary memory rework)
- **Adhesive Dispense Equipment:** Able to dispense thermal cure edge bond adhesive on memories (within 1mm KOZ) and on the SoC (Example: a piezo electric jetting valve with 200um ID nozzle)
- **Height Sense Feature:** Dispense equipment has a height sense feature to create a zero datum for the Z-axis (targeting the Intel SoC)
- As different memories have different Z-heights, need to ensure dispense nozzle maintains $\geq 200\mu\text{m}$ spacing from the top of each memory height
- **Heated nozzle:** Example: 60°C , to allow for a lower viscosity to achieve smaller dot weights / more accurate dispense
- **Adhesive (edge bond) material:** That performs well in Temperature Cycling SJR Testing (example: Zymet UA-2605B* or Zymet UA-2625B*).
- Any adhesive used must be reworkable & removeable, without causing damage to the Intel SoC, for returns or FACR purposes

Expected Outcome:



Ensure No Insufficient Volume of Glue Between Memories that prevents Full Wetting

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Reflow Oven, to cure
the edge bond
adhesive

Minimum Rqmt's:

- **Basic Reflow Oven:** Capable to cure the edge bond adhesive at a low curing temperature, per the adhesive manufacturer's specification (example: Zymet UA-2605-B * cures at 140C for 5 minutes)
- No Nitrogen requirement

Expected Outcome:

- Reflow oven completely cures the edge bond material that is applied on the memory and SoC

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Razor Lake-Ax MoP Equipment List

BGA Rework Equipment, to rework a memory package(s)

Minimum Rqmt's:

- **Automated rework tool with optical BGA alignment capabilities:** Nitrogen capable (N2 turned on), with both top-side & bottom-side heating capabilities, and a topside heating nozzle
- **Top-side Heating Nozzle Sizes:**
 - Nozzle #1-to rework one memory (size 9x15mm)
 - Nozzle #2-to rework two memories (size 9x30mm)
 - Nozzle #3-to rework three memories (size 9x45mm)
 - Nozzle #4-to rework four memories (size 9x55mm)
- **Hot air pencil:** To apply heat used to remove the memory adhesives
- **3mm Torlon* scraper:** Or another tool that does not cause any SoC soldermask damage. Light force is applied when removing memory adhesives
- **Memory Shielding Materials:** Examples: Chemtronics* CM8 Chemask® or Techspray* Wondermask P 2211®, to protect (provide a thermal barrier) the "good" memories not being reworked, dispensed with either a spatula or syringe
- **Cure Oven:** To cure the memory shielding material
- **Rework Pallet Top & Bottom:** Example: Aluminum material, to provide robust motherboard support to minimize PCB warpage during rework
- **Fine-Braided Solder Wick, Flux, Wide-tip Soldering Iron:** For site-dressing
- **Flux Material:** The same flux material used during the SMT MoP process, manually dispensed from a syringe, then spread with a small brush and plastic spatula.

Expected Outcome:

- Successful memory reworkability with no SJ damage on SLI or non-reworked memories

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Razor Lake-Ax MoP Equipment List

Enhanced Memory
Dye and Pull Process,
to remove it from the
top of the SoC

Minimum Rqmt's:

- **Pulling Tool MTS Criterion Model 43***: With a fixture and spacer for customization
- **A thick backing plate attached to the PCB sample**: Especially important for thin motherboards to improve pull results
- **Dremel 4300* Hand Polisher**: To make the surface rough
- **Red Dye**: To highlight any solderjoint cracks
- **Vacuum Chamber**: Used for the red dye penetration into potential BGA crack lines
- **MIXPAC* Dispenser Gun**: Used to control the epoxy dispense
- **3M* Epoxy 460**
- **3M* 501+ High Temperature Masking Tape**
- **Pull Nut**: The nut size should be close to the memory size
- **Cure oven**: Used to cure the epoxy material

Expected Outcome:

- Successful DnP operation of the memory (off the top of the SoC), where 80-90% of the memory solderjoints can be inspected for crack type & crack size

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